

# Mo Sha

## Ph.D. Candidate in Computer Science

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## PUBLICATIONS

- **Self-adaptive Graph Traversal on GPUs.**  
M Sha, Y Li, and KL Tan. [PDF] [Video1] [Video2]  
International Conference on Management of Data (SIGMOD), 2021.
- **Accelerating Depth-First Traversal by Graph Ordering.**  
Q Lyu, M Sha, B Gong, and K Lyu. [PDF]  
Scientific and Statistical DB Management Conference (SSDBM), 2021.
- **GPU-Accelerated Subgraph Enumeration on Partitioned Graphs.**  
W Guo, Y Li, M Sha, et al. [PDF]  
International Conference on Management of Data (SIGMOD), 2020.
- **GPU-based Graph Traversal on Compressed Graphs.**   
M Sha, Y Li, and KL Tan. [PDF] [Code] [Video] [Slides]  
International Conference on Management of Data (SIGMOD), 2019.
- **Accelerating Dynamic Graph Analytics on GPUs.**  
M Sha, Y Li, B He, and KL Tan. [PDF] [Code] [Video] [Slides]  
International Conference on Very Large Data Bases (VLDB), 2018.
- **Parallel Personalized Pagerank on Dynamic Graphs.**  
W Guo, Y Li, M Sha, and KL Tan. [PDF]  
International Conference on Very Large Data Bases (VLDB), 2018.
- **River: A Real-time Influence Monitoring System on Social Media Streams.**  
M Sha, Y Li, Y Wang, W Guo, and KL Tan. [PDF]  
IEEE International Conference on Data Mining (ICDM), 2018.
- **Heterogeneous Embedding Propagation for Large-scale E-Commerce User Alignment.**  
VW Zhang, M Sha, Y Li, et al. [PDF]  
IEEE International Conference on Data Mining (ICDM), 2018.

## PATENT

- **Cross-section extraction for vortex detection.**  
M Sha, J Liu, Y Ke, E Leonardi, and K Menzies.  
Rolls-Royce PLC, US20190303515A1/EP3547174A1, filed on Mar 2019.

## HONORS & AWARDS

- Dean's Graduate Research Excellence Award, NUS, 2020.
- Research Achievement Award, NUS, 2018.
- Gold Medal, ACM/ICPC Asia Regional, ACM, 2012.
- The CCF Outstanding Undergraduate Award, CCF, 2012.

## PROFESSIONAL SERVICES

- **External Reviewer:**
  - International Conference on Very Large Data Bases, 2021.
  - IEEE International Conference on Data Engineering, 2019.
- **Invited Reviewer:**
  - Transactions on Knowledge and Data Engineering, 2021.
  - Journal of Parallel and Distributed Computing, 2019.
  - IEEE Access, 2019.

## TEACHING

- **Teaching Assistant:**
  - CS5234: Algorithm at Scale, NUS, 2019.
  - CS3223: Database Systems Implementation, NUS, 2018-2021.
  - CS1101S: Programming Methodology, NUS, 2018.

## EDUCATION

Ph.D. in Computer Science

supervised by: Prof. Kian-Lee Tan

**National University of Singapore**

Aug 2017 – 2021 (expected) Singapore

B.Eng. in Computer Science

**Harbin Institute of Technology**

Sep 2010 – Jun 2014 Harbin, China

## EXPERIENCES

Research Assistant

**Database Lab, School of Computing, NUS**

Jun 2016 – Jun 2017 Singapore

Our team was interested in dynamic graph analytics on various applications and was exploring to utilize different hardware hierarchies, e.g., GPU accelerators and their combined architecture, for higher efficiency.

Project Officer

**Rolls-Royce@NTU Corporate Lab, NTU**

Jun 2015 – Jun 2016 Singapore

I took part in the research project focusing on the automatic vortex extraction and tracking in CFD data, which is an essential tool for the aerodynamic design of Rolls-Royce's jet engine.

Algorithm Engineer

**Alibaba**

Jun 2014 – Jun 2015 Beijing, China

I was responsible for the backend of the business logic layer of internet advertisement delivery engine, which was mainly related to user profiling and targeted advertisement indexing, retrieving and recalling.

Software Engineering Intern

**Google**

Jun 2013 – Nov 2013 Shanghai, China

Our team was in charge of developing the backend transaction logic management of the Demand Side Platform, and my main contribution during the internship was the RetryAction module.